

Foundation of Blockchain Technology (IS501) Pre-Reading Material – Lecture 22

Smart Contract

Estimated Reading Time: 15–20 minutes

PRE-READING: SMART CONTRACTS

What is a Smart Contract?

A **smart contract** is a piece of computer code stored on a blockchain that can automatically execute predefined actions when specified conditions are satisfied.

In simple words:

Smart Contract = Rules written as computer code + Automatic execution on blockchain

The basic idea is:

IF condition is satisfied → THEN perform an action

For example:

IF a borrower completely repays a loan
THEN release the collateral.

Smart contracts are an important part of Ethereum because Ethereum was designed not only for cryptocurrency but also for **programmable decentralized applications (DApps)**.

Why Do We Need Smart Contracts?

In a traditional system, an agreement may require a **bank, lawyer, broker, or other intermediary** to verify conditions and perform an action.

For example:

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Course Repository: <https://github.com/rajesh101191/Foundation-of-Blockchain-Technology>

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Traditional process:

Buyer → Intermediary → Seller

With a smart contract, predefined rules can be written into code:

IF required conditions are satisfied



Execute the action



Record on blockchain

This can reduce the need for manual intervention and help automate transactions. Ethereum's financial applications use smart contracts for processes such as loan repayment and escrow.

Simple Example

Imagine a **digital vending machine**.

You insert money → select a product → the machine checks the conditions → the product is released.

A smart contract follows a similar idea:

IF payment is received



Check condition



Execute programmed action

For example, in a supply-chain application:

IF goods are delivered

AND delivery is verified



Release payment

This type of automated execution is one of the important uses of smart contracts.



Smart Contract and DApp

Two terms you should know:

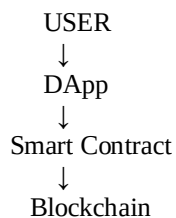
Smart Contract:

The blockchain-based program containing the rules and logic.

DApp (Decentralized Application):

An application that interacts with blockchain and smart contracts.

A simplified structure is:



How Do We Write Smart Contracts?

On Ethereum, smart contracts are commonly written using **Solidity**.

Example:

```
contract Example {  
    string public message = "Hello World";  
}
```

Here:

- contract → defines a smart contract
- Example → name of the contract
- string → data type
- message → variable
- "Hello World" → stored value

You will learn these concepts step-by-step in the next classes.

Key Points to Remember

Before coming to class, remember these **five points**:

1. A smart contract is **computer code stored on a blockchain**.
2. It contains **predefined rules and conditions**.
3. It can **automatically execute actions** when conditions are satisfied.
4. Smart contracts are widely used in **Ethereum-based DApps**.
5. **Solidity** is a programming language used to develop Ethereum smart contracts.

Think Before Class 🧐

1. If a smart contract is computer code, who executes the code?
2. What happens if the condition written in a smart contract is never satisfied?
3. How is a smart contract different from a normal contract written on paper?
4. Can you think of one real-life process that could be automated using a smart contract?

One-Line Takeaway

A smart contract is a blockchain-based program that automatically executes predefined rules when specified conditions are met.

